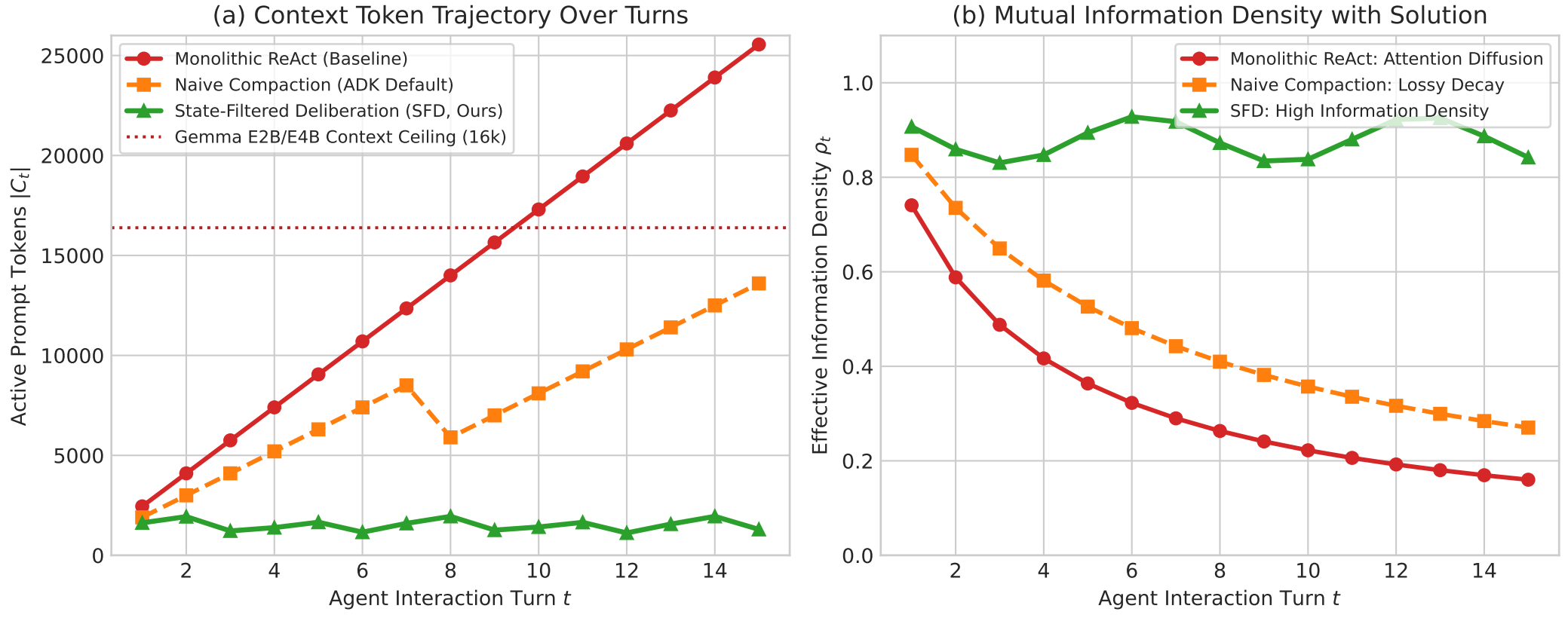
(a) Context Token Trajectory Over Turns
Monolithic ReAct (Baseline)
Naive Compaction (ADK Default)
State-Filtered Deliberation (SFD, Ours)
Gemma E2B/E4B Context Ceiling (16k)
Active Prompt Tokens $|C_t|$
Agent Interaction Turn t
0
5000
10000
15000
20000
25000
2
4
6
8
10
12
14
(b) Mutual Information Density with Solution
Monolithic ReAct: Attention Diffusion
Naive Compaction: Lossy Decay
SFD: High Information Density
Effective Information Density ρ_t
Agent Interaction Turn t
0.0
0.2
0.4
0.6
0.8
1.0
2
4
6
8
10
12
14